



Global, high-end antenna manufacturers

They support multi-band, multi-constellation GNSS (L1/L2/L5, L-Band) for RTK, PPP, and precision applications. Here are 5 well-established brands used in surveying, autonomous systems, agriculture, robotics, and defence-grade positioning:

1. Taoglas. Taoglas is widely integrated into embedded systems and custom electronics designs.

Headquarters: Ireland

Strength. High-performance GNSS, multi-band, multi-constellation antennas. Strong portfolio for RTK and PPP applications, L1/L2/L5 + L-Band support, designed for urban canyon and harsh industrial environments. Popular in robotics, telematics, UAVs, and precision IoT

2. Tallysman wireless. Very respected in high-accuracy surveying markets.

Headquarters: Canada

Strength. Survey-grade GNSS antennas. Excellent phase centre stability (critical for RTK accuracy), multi-frequency GPS, Galileo, BeiDou, GLONASS support, known for low axial ratio and strong multipath rejection, used in precision agriculture, reference stations, and UAV mapping

3. NovAtel (Part of hexagon AB). NovAtel is frequently used in advanced robotics and autonomous system R&D.

Headquarters: Canada

Strength. Offers both high-end GNSS receivers and precision antennas. Common in autonomous vehicles, defence, and heavy machinery, strong RTK and PPP integration, known for industrial-grade robustness

4. Trimble Inc. Trimble antennas are widely deployed in heavy construction automation.

Headquarters: United States

Strength. Premium survey and machine-control GNSS antennas. Industry leader in construction and precision agriculture, advanced RTK and PPP correction ecosystem (RTX services), high reliability in extreme environments

5. Septentrio. Excellent for challenging RF environments.

Headquarters: Belgium

Strength. Known for advanced anti-multipath and anti-jamming performance. Multi-frequency, multi-constellation GNSS, strong interference mitigation technology, used in autonomous vehicles, marine, and defence systems

Other premium GNSS antenna brands worth considering:

- Topcon Corporation
- Leica Geosystems (also part of Hexagon)
- u-blox (strong in modules; antennas via partners)
- Maxtena (compact high-performance designs)

three-centimetre improvement in grading accuracy can deliver substantial savings across large infrastructure projects.

For wider coverage without deploying local base stations, precise point positioning (PPP) and PPP-RTK hybrid solutions provide practical alternatives. Companies such as Hexagon AB and u-blox offer satellite-based correction services that enable decimetre- to centimetre-level accuracy worldwide.

In precision agriculture, autonomous machinery must repeat planting paths consistently across thousands of hectares. PPP-RTK services allow sub-2cm repeatability without installing base stations on large rural properties. Selecting a correction method requires considering convergence time, baseline distance, atmospheric effects, and support for multiple constellations,

including GPS, Galileo, BeiDou, and GLONASS.

Long baselines in mountainous regions can reduce RTK accuracy, unless network RTK services or ionospheric models are used. In offshore energy operations, where base stations are impractical, PPP enables centimetre-level positioning

for survey vessels, supporting subsea cable installation, and offshore wind farm alignment.

Connectivity and validation in the field

In laboratory conditions, centimetre-level GNSS performance appears relatively straightforward. In operational environments, however, connectivity and validation often determine overall success.

RTK corrections are typically delivered through NTRIP over 4G or 5G networks.

In dense urban environments, such as Tokyo and San Francisco, signal blockages and cellular dead zones can interrupt correction streams used by delivery robots and mapping vehicles. Developers address these challenges through redundant SIM cards, network aggregation, and backup positioning modes that combine GNSS with IMUs and visual odometry.

Precision rail monitoring systems in Europe demonstrate how GNSS integrates with track inspection platforms to detect rail geometry deviations. Even short RTK outages can reduce measurement confidence. To maintain reliability, these systems rely on buffered corrections and real-time integrity monitoring.

Validation procedures are equally important.

In UAV corridor mapping for power line inspection, operators fly calibration routes over surveyed

Implementation considerations for engineers

When deploying high-precision positioning systems, address these practical factors:

- **Firmware and processing optimisation.** Support high update rates (10-20Hz or more) and real-time ambiguity resolution to maintain centimetre accuracy in dynamic conditions.
- **Power management.** Optimise SoC performance, duty cycling, and battery capacity to handle RTK or PPP processing without reducing operating time.
- **RF shielding and PCB design.** Minimise EMI and cross-talk through proper grounding, signal isolation, impedance control, and disciplined RF layout.
- **Correction stream reliability.** Use redundant RTK or PPP correction channels, cellular, satellite, or base stations with buffering and automatic fallback to ensure continuity.
- **Environmental protection.** Shield antennas and sensors from vibration, temperature extremes, dust, and moisture to preserve signal quality.
- **Calibration and alignment.** Perform initial and periodic calibration of IMUs and wheel odometry, and maintain antenna alignment to reduce drift and bias.
- **Error monitoring and diagnostics.** Monitor signal quality, correction status, and sensor health in real time to detect faults early and improve system resilience.